



# A model of speed profiles for traffic calmed roads

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## Abstract

The influence of traffic calming measures on the speed of unimpeded vehicles has been investigated by evaluating differences in speed profiles obtained from various combinations of traffic calming measures. A case study has been conducted in the City of York (UK) focusing on traffic calming measures such as speed humps (flat-topped and round topped), speed cushions and chicanes implemented in sequence. Vehicles' passing times were simultaneously recorded at 16 points along each traffic calmed link. From these data a speed profile for each individual vehicle could be derived. An empirical model was developed using multiple regression analysis techniques based on data collected at three calibration sites. Speeds along these links were described as a function of the input speed, the type of measure and the distance between measures. The speed profile model was shown to be a good representation for the data from the calibration sites. It efficiently predicted speeds of unimpeded vehicles over a given combination of traffic calming measures in sequence. The validation process, based on data collected at three validation sites, also indicated that the model provided a good representation of the observed profiles at these sites, with the exception of the prediction of the effects of the chicanes on speeds. This type of measure was shown to produce diverse impacts on speeds which depended on the detailed design. While the model is a useful design tool, recommendations have been made for further enhancement to it. © 2000 Elsevier Science Ltd. All rights reserved.

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## 1. Introduction

Traffic calming schemes have usually been implemented to reduce overall speed and the number of accidents. Consequently, assessment of schemes has been based solely on these parameters, and aspects equally important to scheme design, such as drivers' behaviour when negotiating traffic calming measures, have been neglected. Although more recently studies have been dedicated to the assessment of the effect of individual traffic calming measures, there is still a lack of information on the relative impact of different measures and different combinations of measures.

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This article presents the results of a detailed analysis of the influence of different combinations of traffic calming measures on the speed of unimpeded vehicles. This has been undertaken by evaluating the differences in speed profiles obtained from various combinations of traffic calming measures involving speed humps (flat-topped and round topped), speed cushions and chicanes implemented in sequence. The emphasis throughout is on the formulation of an empirical model, based on data collected at six sites in the City of York (UK), which explains the variations in speed profiles according to the type and position of measures within a sequence.

## 2. Selected traffic calming schemes

The City of York was identified as a suitable place to undertake the field work as it offers a wide range of traffic calming techniques and a large number of potential survey sites (Durkin and Pheby, 1992). Time, economic constraints and management problems were factors also taken into account in this decision. Five roads were identified for the case study in York: Townend Street in The Groves; Livingstone Street in the Leeman Road Area; Fourth Avenue in the Tang Hall Area; and Gale Lane and Foxwood Lane in Acomb. Two sections of Foxwood Lane were selected, therefore the first section which comprises a sequence of three cushions has been denominated as Foxwood Lane East and the remaining section as Foxwood Lane West (see Fig. 1). All roads are two-way except Livingstone Street and Townend Street.

According to work by Hodge (1992), the most frequent traffic calming measures in the UK are: flat-topped tables, round-topped humps, raised junctions, pinch-points/narrowings, speed cushions and chicanes. These roads in York follow the trends observed by Hodge (1992), but with few raised junctions or individual pinch points. Consequently the survey approach has focused on the following measures: speed humps (flat and round topped), speed cushions and chicanes. The flat topped hump has been termed a ‘table’, the round topped one a ‘hump’, and the speed cushion a ‘cushion’, throughout this article. Three survey sites were used to calibrate a model which has been validated using data from another three survey sites. Table 1 shows the sequence of measures at each site, the measures for which data was collected, and their use for calibration and validation.

Table 1  
Survey sites—type of measures and categorisation

| Livingstone Street   | Foxwood Lane West    | Foxwood Lane East    | Gale Lane            | Townend Street       | Fourth Avenue        |
|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| Junction             | Hump <sup>a</sup>    | Cushion              | —                    | Junction             | Junction             |
| Chicane <sup>a</sup> | Table <sup>a</sup>   | Cushion <sup>a</sup> | Cushion              | Hump                 | Table                |
| Chicane <sup>a</sup> | Cushion <sup>a</sup> | Cushion <sup>a</sup> | Cushion <sup>b</sup> | Table                | Table <sup>b</sup>   |
| Junction             | Cushion              | Cushion <sup>a</sup> | Cushion <sup>b</sup> | Hump                 | Hump <sup>b</sup>    |
|                      |                      | Cushion              | Cushion <sup>b</sup> | Table                | Cushion <sup>b</sup> |
|                      |                      |                      | Cushion              | Chicane <sup>b</sup> | Cushion              |
|                      |                      |                      | Cushion + island     | Chicane <sup>b</sup> | Cushion              |
|                      |                      |                      | —                    | Hump <sup>b</sup>    | Cushion              |
|                      |                      |                      |                      | Junction             | —                    |

<sup>a</sup> Calibration.

<sup>b</sup> Validation.

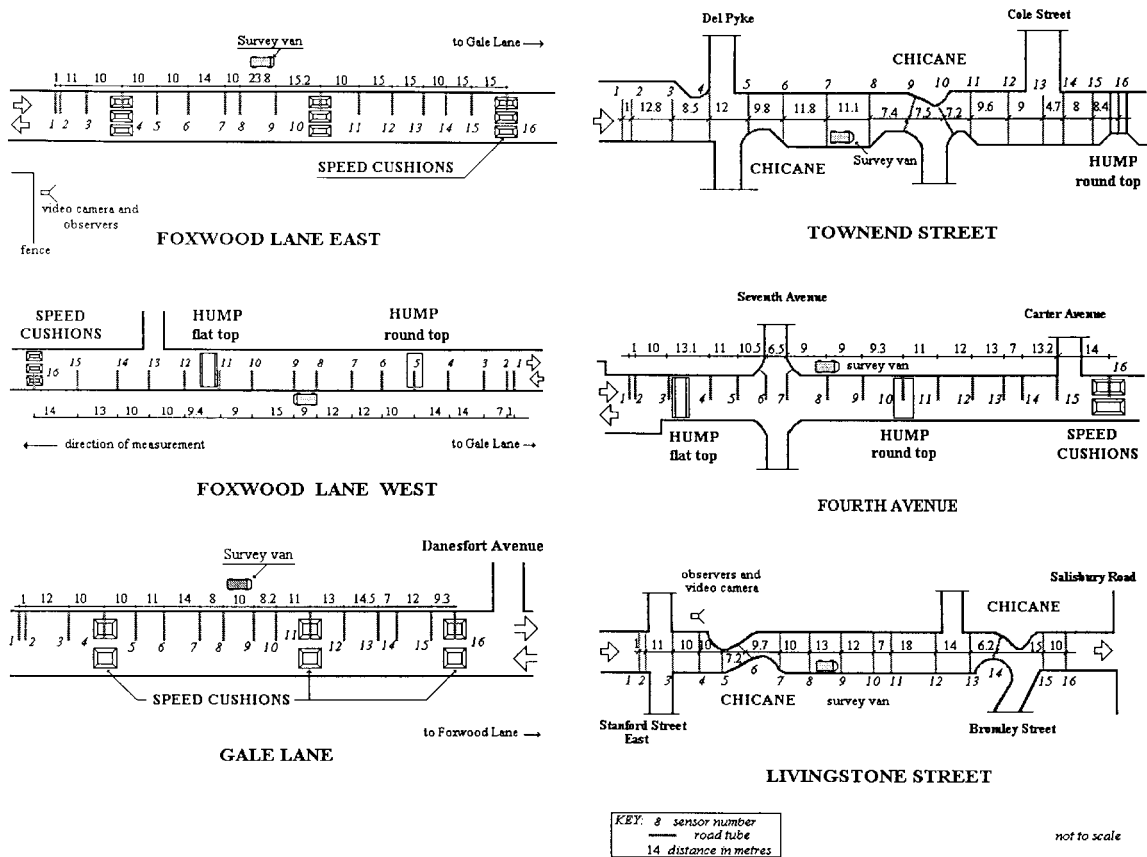


Fig. 1. Plans of study sites.

### 3. Data collection

The main data collected is related to vehicles' speed at predetermined points along traffic calmed links as it is the parameter that best defines drivers' behaviour. Data was collected on weekdays during off-peak hours when there is more likely to be free flow traffic and, consequently, less interaction between vehicles. Impeded vehicles were not of interest, since their speeds are affected by the lead vehicle rather than by the traffic calming measures. The field work was carried out in daylight and, where possible, in good weather conditions in order to avoid variations in drivers' behaviour due to different weather conditions.

Many techniques are available for the measurement of vehicle speeds but few can easily provide this data at various points along a selected link. A pilot test indicated that a data logger (plus associated equipment) was the most suitable for this study. In addition to data recorded automatically, complementary data would also be required, especially for data selection and editing. The chosen approach involved the use of observers in conjunction with a video camera.

### 3.1. *Data collection system*

The automatic system of recording events was based on a data logger computer working together with a portable computer and connected to 16 automatic sensors through 16 input channels. The sensors consisted of pneumatic tubes fixed onto the road, each connected to an air-switch which transforms the air pulse into electric signals. This signal was sent to the data logger through an input/output board. The network consisted of nodes (boxes containing the air-switches) and links (ribbon type cables). Each node (air-switch box) was connected to the network through a water-proof connector which allowed some flexibility to change the position of the connections according to the distance between tubes for each site. The data acquisition system recorded vehicle axle passing time rather than vehicle speed. Data logger, stop watches and video camera were synchronised in order to get consistent sets of data. Fig. 1 shows the location of the traffic calming devices and sensors for each of the six sites.

It was recognised that 16 sets of pneumatic tubes may in themselves influence driver speeds, but tests showed (Barbosa, 1995) that although there was an initial reduction in mean vehicle speeds as a result of the tubes, this effect was reduced over time. Thus measurements were only taken once the equipment had been in place for 3 days.

### 3.2. *Data selection*

Since the target vehicles were restricted to those which are typical of a traffic calmed road, only cars and light goods vehicles (vans) were considered in the sample while buses, coaches and other goods vehicles were excluded. For the purpose of this study, the term ‘van’ is applied to all light goods vehicles except lorries according to the UK Department of Transport (DET/IHT, 1987) classification. As this study is an investigation of drivers’ behaviour when they are negotiating calming measures, it is reasonable to study behaviour when a driver has absolute control over the decision about his/her driving style without being pressured by other vehicles. A vehicle travelling under these conditions has been termed an ‘unimpeded vehicle’, which implies that the driver is not being affected by the motion of the vehicle immediately preceding him/her in the traffic stream or by other agents (pedestrians, cyclists, opposing vehicles). The minimum gap to determine whether a preceding car is affecting speed was defined for each site as the headway which allows a spacing equal to the distance between traffic calming measures along the link. In the light of these definitions, the following data were removed: (a) vehicles starting or stopping within the street section; (b) vehicles others than cars or vans; (c) vehicles whose behaviour was affected by pedestrians, cyclists or animals; (d) vehicles in conflict with opposing vehicles; (e) vehicles for which there was incomplete data; and (f) vehicles less than the specified headway from the preceding vehicle.

### 3.3. *Speed profile calculation*

Road tube layout varied from road to road but the first pair of tubes at the entrance to the measured section were always placed one metre apart. Passing time data obtained from this pair of tubes was crucial to determine the distance between axles of vehicles (wheelbase). A spacing of 1 m was adopted in order to reduce the possibility of vehicles accelerating or decelerating in between the tubes. The calculation of the wheelbase was carried out as follows:

- a. Two approximate spot speeds are obtained:  $V_1$  from the difference of passing times of the first axle over sensors 1 and 2 and  $V_2$  from the difference of passing times of the second axle over the same sensors.
- b. Still using the same pair of sensors (1 and 2), four estimates of the length of the wheelbase can be obtained from the difference between passing times of the first and second axles over sensor 1 ( $T_1$ ) and sensor 2 ( $T_2$ ), multiplied by the spot speed ( $V_1$  and  $V_2$ ) previously calculated. As demonstrated by Seco (1991), a good estimate of the wheelbase value can be obtained by simply averaging the values  $V_1T_1$  and  $V_2T_2$ .

Once the wheelbase has been determined the speed profile for each individual vehicle is achieved as a result of calculating the spot speed on the other 14 single sensors by dividing the wheelbase by the difference of passing times of the first and second axle over each sensor.

The completion of data processing for the six survey sites resulted in a database of sample size shown in Table 2 and for each selected vehicle a speed profile containing 16 speed values along the survey link was achieved.

#### 4. Data analysis

The method of speed data collection on traffic calmed links involving a sequence of measures, results in a relationship between speed and distance which has been termed a *speed profile*. Table 3 presents the cumulative distances in respect of each data point for each survey site. In addition, the actual positions and types of traffic calming measure are also indicated.

For each site, data were allocated to seven speed classes based on entry speed to the site (apart from Foxwood Lane West, where six were used); the size of the classes was determined by the distribution of entry speeds. Figs. 2–4 present the speed profiles of individual vehicles plotted for a sample of the survey sites according to an entry speed class which contains the average entry speed ( $V_1$ ) observed at each site (Foxwood Lane West: 36.61 km/h, Foxwood Lane East: 31.79 km/h and Livingstone Street: 28.75 km/h). The complete set of speed profile plots for all entry speed classes is presented in Barbosa (1995).

The overview of speed profiles outlined differences and similarities among the surveyed links and provided some information on drivers' behaviour. There is a remarkable similarity among

Table 2  
Resulting database for each survey site

| Sites              | Database size |      |       |
|--------------------|---------------|------|-------|
|                    | Cars          | Vans | Total |
| Livingstone Street | 288           | 70   | 358   |
| Foxwood Lane East  | 317           | 27   | 344   |
| Foxwood Lane West  | 138           | 15   | 153   |
| Gale Lane          | 286           | 26   | 312   |
| Townend Street     | 210           | 30   | 240   |
| Fourth Avenue      | 146           | 19   | 165   |
| Totals             | 1385          | 187  | 1572  |

sites in terms of driving style: the higher the entry speed the less the speed reduction. The entry speed is usually the maximum speed along the link and may explain a proportion of the variance in driver behaviour, but may also be partially explained by the characteristics of the link itself. The most remarkable difference between links is associated with the general shape of speed profiles. The three different measures at Foxwood Lane West (hump, table and cushion) (Fig. 2) have produced speed profiles which highlight the peaks and troughs, with the troughs indicating the

Table 3  
Cumulative distance (in m), position and type of measures—all sites

| Data points | Foxwood Lane West   | Foxwood Lane East | Livingstone Street | Fourth Avenue | Gale Lane | Townend Street |
|-------------|---------------------|-------------------|--------------------|---------------|-----------|----------------|
| 1           | 0.0                 | 0.0               | 0.0                | 0.0           | 0.0       | 0.0            |
| 2           | 1.0                 | 1.0               | 1.0                | 1.0           | 1.0       | 1.0            |
| 3           | 8.0                 | 12.0              | 12.0               | 11.0 T        | 13.0      | 13.8           |
| 4           | 22.0                | 22.0 C            | 22.0               | 24.1          | 23.0 C    | 22.3           |
| 5           | 36.0 H <sup>a</sup> | 32.0              | 32.0               | 35.1          | 33.0      | 34.3 Ch        |
| 6           | 46.0                | 42.0              | 39.2 Ch            | 45.6          | 44.0      | 44.1           |
| 7           | 58.0                | 56.0              | 48.9               | 52.1          | 58.0      | 55.9           |
| 8           | 70.0                | 66.0              | 58.9               | 61.1          | 66.0      | 67.0           |
| 9           | 79.0                | 89.8              | 71.9               | 70.1          | 76.0      | 74.4           |
| 10          | 94.0                | 105.0 C           | 83.9               | 80.1 H        | 84.2      | 81.9 Ch        |
| 11          | 103.0 T             | 115.0             | 90.9               | 91.1          | 95.2 C    | 89.1           |
| 12          | 112.4               | 130.0             | 108.9              | 103.1         | 108.2     | 98.7           |
| 13          | 122.4               | 145.0             | 122.9              | 116.1         | 122.7     | 107.7          |
| 14          | 132.4               | 155.0             | 129.1 Ch           | 123.1         | 129.7     | 112.4          |
| 15          | 145.4               | 170.0             | 144.1              | 136.3         | 141.7     | 120.4          |
| 16          | 159.4 C             | 185.0 C           | 154.1              | 151.3 C       | 151.0 C   | 128.8 H        |

<sup>a</sup> T = table; H = hump; C = cushion; Ch = chicane.

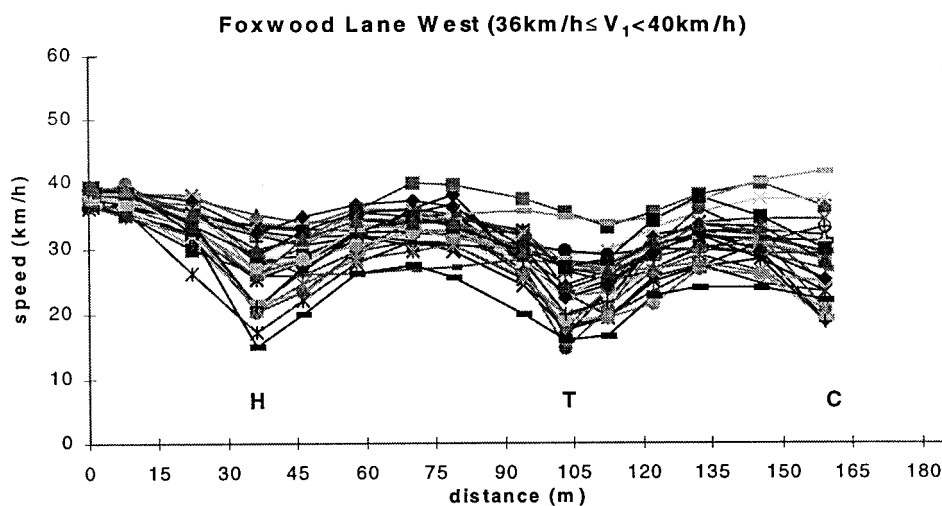


Fig. 2. Speed profile plots for Foxwood Lane West ( $36 \leq V_1 < 40$  km/h).

severity of the measures. The differences in shape become more evident when compared to the relatively flat profiles from the three speed cushions in Foxwood Lane East (Fig. 3). Chicanes in Livingstone Street show a distinct profile with lower acceleration and deceleration rates and the highest peaks among all sites (Fig. 4).

The analysis of the speed distribution at the measures has suggested different impacts according to the type of measure. Fig. 5 presents the speed distribution for the calibration sites. Speed profile plots have enabled the visual examination of the slopes which indicate the acceleration rates between a pair of data points. The slopes also suggest different acceleration rates for different types of measure, reinforcing the differences in impact between types of measure.

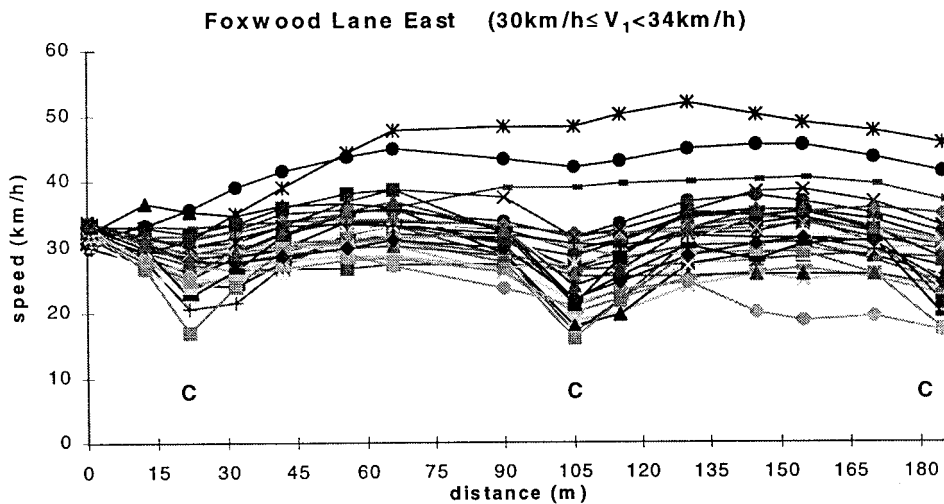


Fig. 3. Speed profile plots for Foxwood Lane East ( $30 \leq V_1 < 34$  km/h).

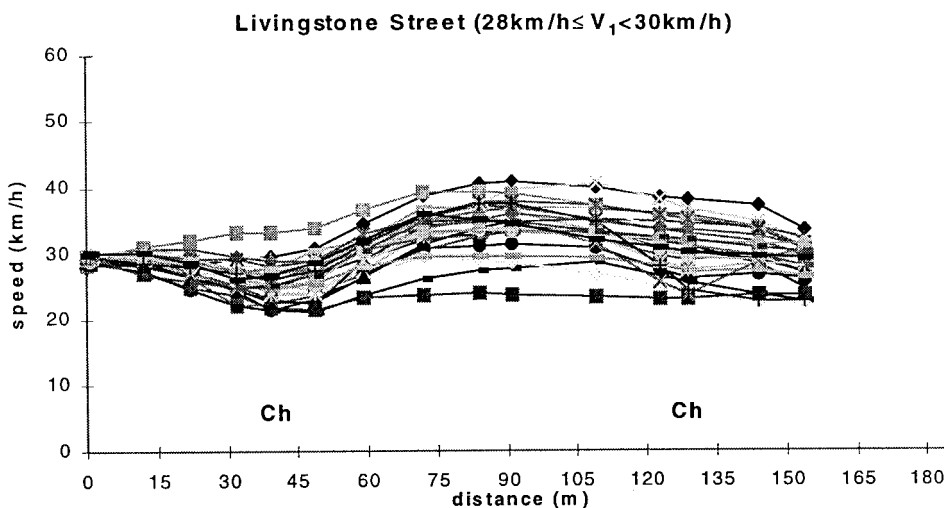


Fig. 4. Speed profile plots for Livingstone Street ( $28 \leq V_1 < 30$  km/h).

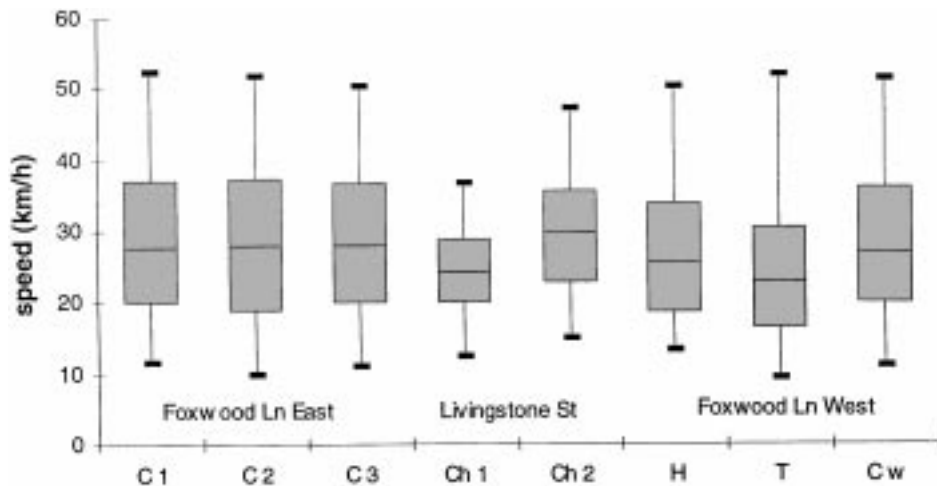


Fig. 5. Minimum, 15th percentile, median, 85th percentile and maximum speeds—calibration sites.

Comparisons among speed profiles at different sites suggest that more aggressive and constraining measures reduce speeds to more desirable levels. Hence, it appears that speed profile curves can be explained by variables such as the entry speed, the type of measure and the distance between devices.

Tests were made of hypotheses involving the consistency of the impacts of measures on speed, the effects on acceleration and deceleration rates, and the impact on travel times. The complete report of those tests is given elsewhere (Barbosa, 1995). This section, however, summarises the main findings which have established the basis for the model development.

Considering the consistency of impacts, it was found that: (a) calming measures of the same design are likely to produce similar impacts on speeds; (b) there was no evidence for measures in sequence producing an additive effect on speeds; measures of the same type and design have a similar impact on speeds wherever they occur in a sequence; (c) the effectiveness of the measures in reducing speed decreases at higher entry speeds; (d) the speed at the device (minimum speed) is mainly affected by the type (geometry) of the device itself, while the distance from the previous measure (which determines the position of maximum speed between measures), and the maximum speed itself play a secondary role; (e) maximum speed downstream of a measure is affected by a combination of the speed (minimum speed) at which the upstream device is crossed and the distance between measures; (f) the position at which maximum speed is attained lies between 55 and 64% of the separation between measures in the direction of travel; and (g) the minimum speed occurred at the measure itself.

The analysis of acceleration rates by type of measure has shown different rates according to the type of measure as shown in Table 4. Speed cushions in Foxwood Lane East all presented similar average acceleration and deceleration rates indicating similar impacts. However, the cushion in Foxwood Lane West indicates less deceleration than the remainder of the cushions (in Foxwood Lane East). The table and the hump show a similar but much higher deceleration, with the hump being higher. The chicanes (Livingstone Street) present similar decelerations



Table 4  
Mean acceleration and deceleration rates by type of measure

| Foxwood Lane West |                            | Foxwood Lane East |                            | Livingstone Street |                            |
|-------------------|----------------------------|-------------------|----------------------------|--------------------|----------------------------|
| Section           | Acc/dec ( $\text{m/s}^2$ ) | Section           | Acc/dec ( $\text{m/s}^2$ ) | Section            | Acc/dec ( $\text{m/s}^2$ ) |
| 3–5               | –0.824                     | 2–4               | –0.345                     | 3–6                | –0.349                     |
| 5                 | Hump                       | 4                 | Cushion                    | 6                  | Chicane                    |
| 5–8               | 0.377                      | 4–7               | 0.297                      | 6–10               | 0.502                      |
| 9–11              | –0.738                     | 8–10              | –0.304                     | 10–14              | –0.286                     |
| 11                | Table                      | 10                | Cushion                    | 14                 | Chicane                    |
| 11–14             | 0.529                      | 10–13             | 0.237                      | 14–16              | –0.219                     |
| 14–16             | –0.253                     | 14–16             | –0.315                     | –                  | –                          |
| 16                | Cushion                    | 16                | Cushion                    | –                  | –                          |

and their values are within the range of acceleration/deceleration induced by the cushions (Foxwood Lane East), while the acceleration from the first chicane is similar to the hump. Vehicles continued to decelerate from the second chicane, indicating the influence of the downstream junction on speed.

Linear relationships established through regression analysis between entry speed ( $V_1$ ) and the total travel time in the surveyed links provided a reasonable degree of explanation for the total travel time. However, the applicability of such relationships is restricted to traffic calming schemes featuring similar combinations of measures to those in the surveyed links. The likely delay (increase in travel time) was determined according to the type of measure by establishing the participation of each measure in the total travel time observed in the links. The area of influence of each measure was taken to be between the locations of two consecutive maximum speeds so that it includes the deceleration and acceleration phases surrounding a measure. Table 5 provides some comparisons of measures by their relative slowness ( $\text{s/km}$ ) and space mean speeds across these areas of influence, and shows that the table measure constrained speeds to a lower level imposing the greatest delay, while cushions showed the highest space mean speeds, hence imposing the least delay.

Table 5  
Travel time by type of measure

| Sites              | Measures | Travel time (s) | Section length (m) | Slowness ( $\text{s/km}$ ) | Space mean speed ( $\text{km/h}$ ) |
|--------------------|----------|-----------------|--------------------|----------------------------|------------------------------------|
| Foxwood Lane West  | Hump     | 8.85            | 70.0               | 126.43                     | 28.47                              |
|                    | Table    | 8.76            | 62.4               | 140.38                     | 25.64                              |
|                    | Cushion  | 3.46            | 27.0               | 128.15                     | 28.09                              |
| Foxwood Lane East  | Cushion  | 7.98            | 66.0               | 120.91                     | 29.77                              |
|                    | Cushion  | 10.52           | 89.0               | 118.20                     | 30.46                              |
|                    | Cushion  | 3.54            | 30.0               | 118.00                     | 30.51                              |
| Livingstone Street | Chicane  | 12.08           | 90.9               | 132.89                     | 27.09                              |
|                    | Chicane  | 7.83            | 61.0               | 128.36                     | 28.01                              |

## 5. The speed profile model

### 5.1. First modelling attempts based on known models

The first attempt to model the data from the speed profile curves used the model suggested by Jarvis (1987) and Pitcher (1989) which relates speed to distance and is of the form:  $V^3 = a + bx + cx^2$  (where  $V$  is velocity,  $x$  is distance and  $a, b$  and  $c$  are constants). Correlation coefficients obtained for models dividing the link into sections and based on the whole sample were very poor, less than 25% for Foxwood Lane West and less than 6% for Foxwood Lane East. When considering speed collected at all data points the results worsened and correlation coefficients of around 5% were achieved when average speed was used. The shape of the observed speed profile curves suggested the use of polynomial models. Although it is advisable to keep the order of the polynomial model as low as possible, an attempt was made to fit a curve which included all 16 data points using a polynomial function of the fifth-order relating speed to distance. This high-order polynomial accounted for the peaks and troughs resulting from the effect of speed reducing devices but it did not account for the type of measure. Time series analysis was another alternative considered in the modelling process. However, this proved very difficult to apply because the control devices are not uniformly spaced within or between sites (as would normally be the case with time-based sequences), and because the nature of the event (i.e. the control device) at a given point in the sequence differs between sites.

### 5.2. Model development

Since the known models failed to explain the speed profile curves, it was decided to develop a model based on the knowledge acquired from the analysis of the data collected. The main intention was to end up with a simple and practical relationship which describes the effects on speed of different measures and would be easily applicable. Regression was the technique adopted for the estimation of parameters in the model.

It is known that the separation of measures influences the maximum speed attained (Lines, 1993; Webster, 1993) and in this particular situation, the position of the calming devices is a major factor determining the variations in speeds. In addition the type of measure was identified as a major factor influencing the speed at the measures (minimum speeds). The separation of measures and their type contribute to the explanation of maximum and minimum speeds, but the maximum speed variation is usually influenced also by the entry speed ( $V_1$ ). Although the entry speed does not represent the free entry speed, as there may be a previous measure constraining speeds, it has been assumed as such throughout this study. This is a simplification which is imposed on the study by the lack of isolated traffic calming measures, which in turn is a direct result of UK DoT (Department of Transport, 1993) regulations.

An objective approach that considers the relative distance between two consecutive measures was devised to describe the position of a traffic calming measure in a link. This was done based on the distance to the next measure (denoted by  $dt$ ) and the distance from the previous measure (denoted by  $df$ ). To transform overall distance into these two variables is a simple operation of subtracting (in case of  $dt$ ) and adding (in case of  $df$ ) the portion travelled along the link in relation to the distance (spacing) between measures, so that  $dt + df = \text{spacing}$  between measures. Zero values of  $dt$  indicate the position of the calming measures.

Dummy variables were used to indicate the presence or absence of calming measures. The effect of the dummy variables was assumed to be restricted to the location where the measure is, that is the effect is solely applied to the data point corresponding to the measure.

Hence, the independent variables considered in the model to describe ‘speed’ are as follows:

- a. *the entry speed* ( $V_1$ ) is the initial speed in the link which was assumed as the speed measured at the first sensor during data collection (noting that this was taken as indicative of free flow speed) (in km/h);
- b. *distance to* ( $dt$ ) is the distance to the next measure in the direction of travel (in m);
- c. *distance from* ( $df$ ) is the distance from the previous measure also measured in the direction of travel (in m);
- d. *hump* ( $H$ ) is the dummy variable which indicates the presence of a hump (0 or 1);
- e. *table* ( $T$ ) is the dummy variable which indicates the presence of a table (0 or 1);
- f. *cushion* ( $C$ ) is the dummy variable which indicates the presence of a speed cushion (0 or 1); and
- g. *chicane* ( $Ch$ ) is the dummy variable which indicates the presence of a chicane (0 or 1).

Simplifications and assumptions were inevitable during the calibration process in order to generate a simple model. In particular, equal impacts were assigned to all measures of a given type and the entry speed ( $V_1$ ) was taken as indicative of free flow speed. The model was developed to consider relatively regular sequences of measures, such as are often encountered on urban residential roads in the UK. Because of this it does not predict speeds well when the gap between two calming devices becomes too large as there is no limiting factor on speed in these conditions.

Multiple linear regression using the ordinary least squares (OLS) method was applied to estimate the parameters of the function which relates the variables specified in the model. However, the residual analysis indicated the presence of heteroscedasticity. The method of weighted least squares was applied to the data, with a modification of the dependent variable, speed, to reduce the degree of heteroscedasticity.

The transformation has been done in two steps: first the usual OLS regression was run disregarding the heteroscedasticity problem to obtain  $\hat{Y}_i$ , the expected value of speed. Then, using the estimated  $\hat{Y}_i$ , the original model was transformed by dividing all terms by the square root of the expected value of speed. Although  $\hat{Y}_i$  are not exactly the expected value of speed, they are consistent estimators; that is, as the sample size increases indefinitely, they converge to the true expected value. Hence, this transformation is valid if the sample size is reasonably large, which is the case.

The recommended speed profile model developed with a weighted least squares regression calibrated for all sites, which has the capability of explaining approximately 55% of the phenomenon ‘speed’,<sup>1</sup> is:

$$\begin{aligned} \text{Speed} = & -8.73 + 0.62V_1 + 0.23dt + 0.78df - 0.0012dt^2 - 0.0137df^2 + 8.5E - 05df^3 \\ & - 4.48H - 6.71T - 0.86C - 2.01Ch \end{aligned}$$

<sup>1</sup> Note that if the values for the entry point are input to the equation the resultant predicted speed at the entry point will not be equal to the observed speed since the input value for  $dt$  will not be zero. However, as this value is known it is felt that this is an acceptable weakness to the model.

The second and third order of  $df$  and  $dt$  are included in the equation because of the shape of the observed speed profile curves which suggest a polynomial function. Speed was investigated as a function of combinations of the partial distance variables,  $dt$ ,  $dt^2$ ,  $dt^3$ ,  $df$ ,  $df^2$ , and  $df^3$ . Analysis showed that the addition of higher order partial distance variables only slightly improved the correlation, however it improved the visual model curve fit substantially.

The correlation coefficients for individual sites were approximately 20% greater than the model calibrated on all sites, for instance that for Foxwood Lane West was around 67%. These relatively low correlations have been attributed to the large variation in the collected speed profiles mainly found near the midpoint between the devices and to differences in driver behaviour, as shown in speed profile plots (Figs. 2–4). The explanatory variable, entry speed, was included to account for a proportion of the variance in driver behaviour. Inevitably, the aggregation of speed data collected at different sites (with different traffic calming measures located at different positions along the links) also contributed to the decrease in the correlation coefficients.

It is worth noting that the coefficients of the last four terms of the equation relating to the different traffic calming measures, give an overall indication of the effectiveness of those measures in reducing speeds (tables are best, followed by humps, chicanes and cushions). A table imposes a reduction of almost 7 km/h, a hump 4.5 km/h, a chicane 2 km/h and a cushion less than 1 km/h.

### 5.3. The model curve fit

This section focuses on the graphical representation of the speed profile model. The curve fits have been provided for the calibration sites. The speed values have been obtained by substituting the partial distance values ( $dt$  and  $df$ ) and the input speed  $V_1 = 33$  km/h (which corresponds to 20 mph, the target speed in calmed areas) into the speed profile model equation. The equation was then used to predict 16 speed values for each site; one for each of the measurement points.

Figs. 6–8 present the modelled speed profile assuming an input speed of  $V_1 = 33$  km/h (thick black line) superimposed on the observed speed profile plots for actual entry speeds.

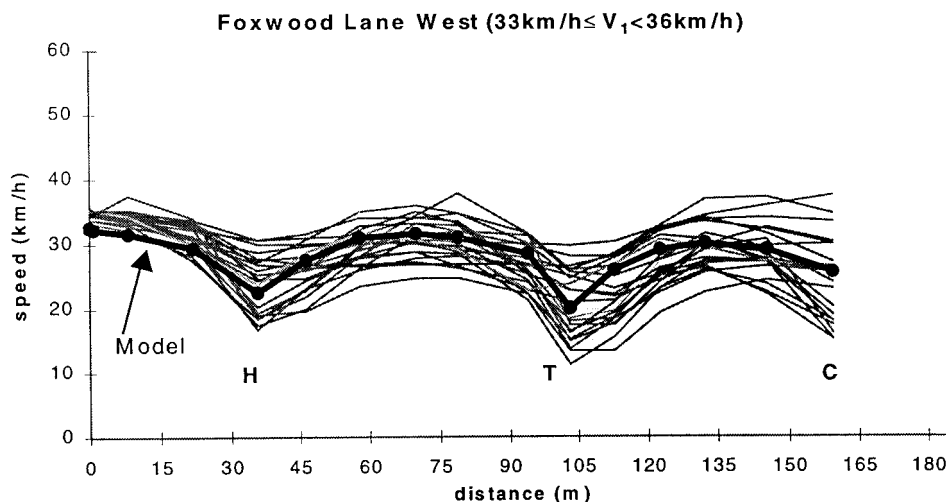


Fig. 6. Modelled speed profile ( $V_1 = 33$  km/h) vs observed speed profiles—Foxwood Lane West.

The modelled speed profile for Foxwood Lane West (Fig. 6) provides a good representation of the general trends: minimum speeds follow the observed average pattern; the entry speed reproduces the speed value (33 km/h) applied to the equation. A reasonable prediction has also been achieved for Foxwood Lane East (Fig. 7), but in this case the predicted entry speed is underestimated. A plausible reason for this low speed may be related to the relatively short distance from the first data point to the first measure, i.e. the first data point does not in practice coincide with the actual free flow speed. The modelled profile indicates more accentuated curves (higher slopes) in the acceleration lengths than the deceleration ones. Minimum speeds appear to follow the average pattern. Nevertheless, maximum speeds occur at data points 7 and 13, whereas the average pattern has indicated similar maximum speed values for data points 7–8 and 13–14.

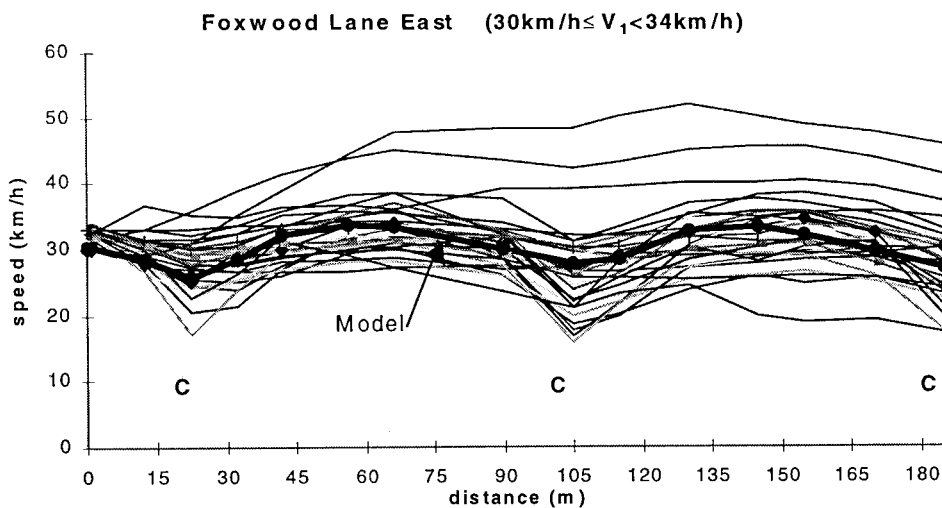


Fig. 7. Modelled speed profile ( $V_1 = 33$  km/h) vs observed speed profiles—Foxwood Lane East.

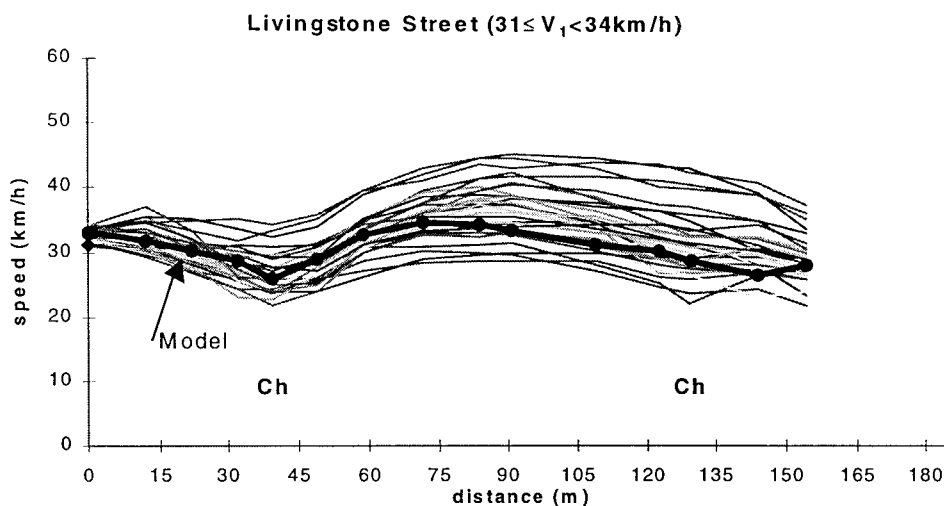


Fig. 8. Modelled speed profile ( $V_1 = 33$  km/h) vs observed speed profiles—Livingstone Street.

Livingstone Street (Fig. 8) also presents a reasonable prediction, except for the last portion of the curve where an atypical deceleration is found after the second chicane. However the portion of the profile which contains the maximum speed between the two chicanes (data point 11) appears flatter than the average profile and indicates that the maximum speed occurs at data point 9 instead.

Input speed values equal to the average speed calculated for the first data point ( $V_1$ ) have also been applied to the model equation, in order to compare average speed profiles against modelled speed profiles. Each site has been associated with a distinct input speed since the average entry speeds vary within the calibration sites. Hence, the input speed for each site is as follows: Foxwood Lane West, 36.61 km/h; Foxwood Lane East, 31.79 km/h; and Livingstone Street, 28.74 km/h. Figs. 9–11 depict the average and the modelled speed profile curves.

The modelled profile for Foxwood Lane West presents the closest representation to the average one. Foxwood Lane East underestimates speeds up to the first chicane; it also underestimates the

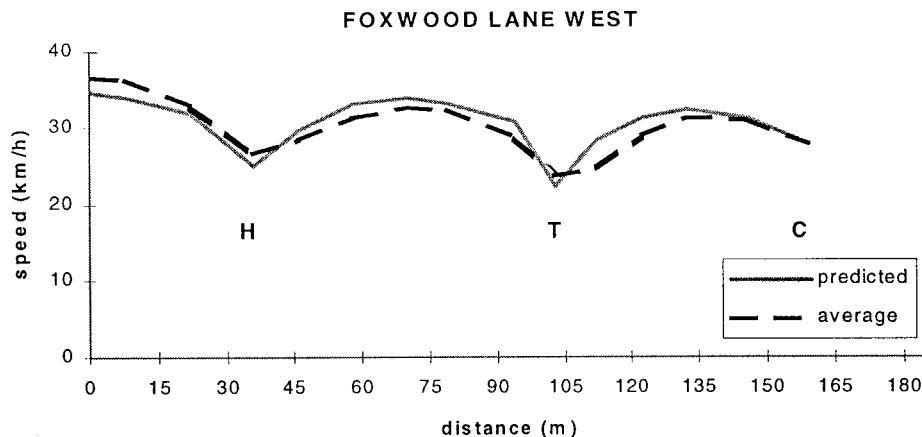


Fig. 9. Average vs modelled speed profile curve—Foxwood Lane West.

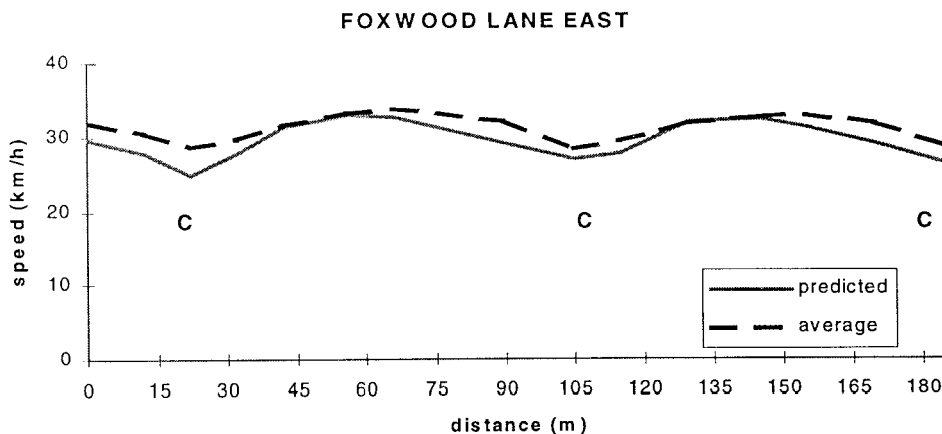


Fig. 10. Average vs modelled speed profile curve—Foxwood Lane East.

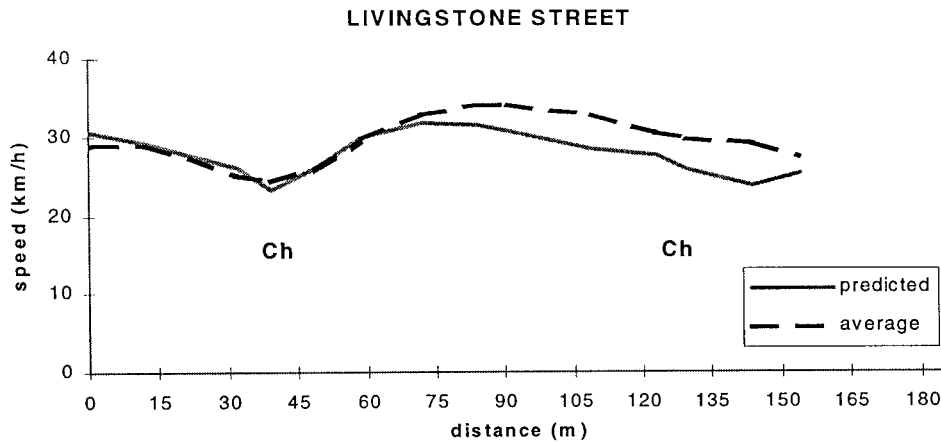


Fig. 11. Average vs modelled speed profile curve—Livingstone Street.

deceleration lengths by transferring maximum speeds one data point earlier than the average profile. The modelled profile for Livingstone Street demonstrates that speeds are being underestimated after the midpoint between chicanes. It appears the predicted profile has been shifted down by a constant value. The last data point presents a discrepancy, caused by the effect of the downstream junction.

#### 5.4. The use of the speed profile model to predict acceleration profiles

Instead of initiating a completely new approach to develop a relationship for acceleration profiles, the speed profile model was the starting point to generate acceleration profiles at calming measures. The use of the difference quotient  $\Delta v/\Delta x$  was preferred to using the first derivative of speed with respect to distance which would have resulted in an unwieldy expression. In this respect the acceleration profile also proved a reasonable prediction tool. It should be stressed that it will reproduce discrepancies encountered in the speed profile model as it was generated from the latter. The general form of the acceleration profile is:

$$\text{acceleration} = \text{speed}(\Delta v/\Delta x)$$

The application of this equation is quite straightforward. Once the values of  $v$  (speed) are known the difference quotient ( $\Delta v/\Delta x$ ) is calculated between each pair of data points, for instance  $\Delta v_{12} = (V_2 - V_1)$ ;  $\Delta x_{12} = (x_2 - x_1)$ . The acceleration profiles obtained are plotted in relation to the midpoint between the data points.

The predicted acceleration profile based on an input speed of 33 km/h has been superimposed on the acceleration profiles of Foxwood Lane West (Fig. 12), for a comparison of trends. The fit is generally good, however, the acceleration rates before and after the table appear overestimated.

#### 5.5. Model validation

The validation process has assessed whether the model adequately represents the data collected at the validation sites: Fourth Avenue, Gale Lane and Townend Street. These sites present some

similarities to the calibration sites concerning the combination of measures encountered at each. As can be seen in Table 3, Fourth Avenue resembles Foxwood Lane West; Gale Lane resembles Foxwood Lane East; and Townend Street resembles Livingstone Street. The speed profile model has been calculated for each validation site adopting an input speed  $V_1 = 33$  km/h. The modelled speed profile, for this input speed, has been examined against the observed profiles for given entry speed ranges surrounding the input speed value. Figs. 13–15 depict the modelled profile (thick black line) superimposed on the observed data.

The predicted entry speeds exhibit lower values than the input entry speeds (33 km/h): Fourth Avenue = 27.79 km/h; Gale Lane = 30.69 km/h; and Townend Street = 30.45 km/h. Fourth Avenue shows the largest difference between predicted entry speeds and input entry speeds, probably as a result of the proximity of the first measure in relation to the link entrance (a situation similar to

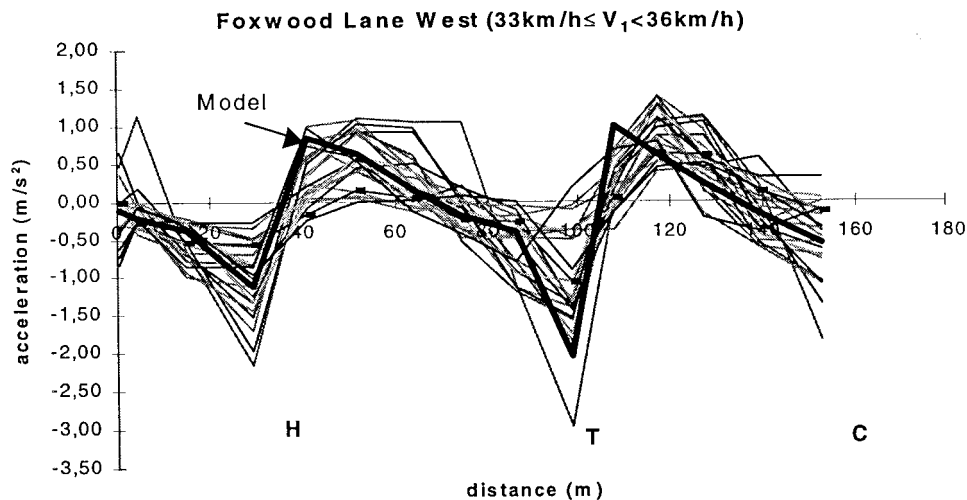


Fig. 12. Predicted acceleration profile ( $V_1 = 33$  km/h) Foxwood Lane West.

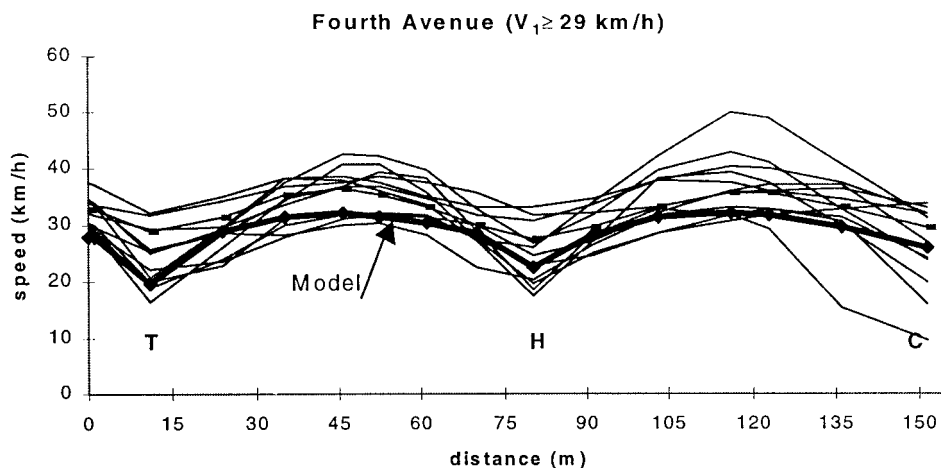


Fig. 13. Modelled speed profile ( $V_1 = 33$  km/h) vs observed speed profiles—Fourth Avenue.



Foxwood Lane East). For Fourth Avenue (Fig. 13), the model follows the observed trends but the speed at the table (the first measure on the link) appears slightly underpredicted. The observed curves suggest more accentuated peaks than the modelled one. However, the observed profiles refer to the highest entry speed range found at this site. The modelled profile achieved for Gale Lane presents a good approximation to the observed curves, but the point of maximum speed between the first and second cushion appears slightly earlier than the actual trends. Contrasting with the above, the model has demonstrated its inadequacy to predict speeds for the two chicanes in Townend Street. This may indicate that these chicanes provoke a different impact on speeds from that of the chicanes used in the model calibration. According to the speed profile curves, the first chicane (at 34.3 m from the origin) induces drivers to maintain or even increase speed rather than reducing it, whereas the model predicts a reduction. The second chicane induces a slight speed reduction, but much less than the model predicts. The hump is the only feature which

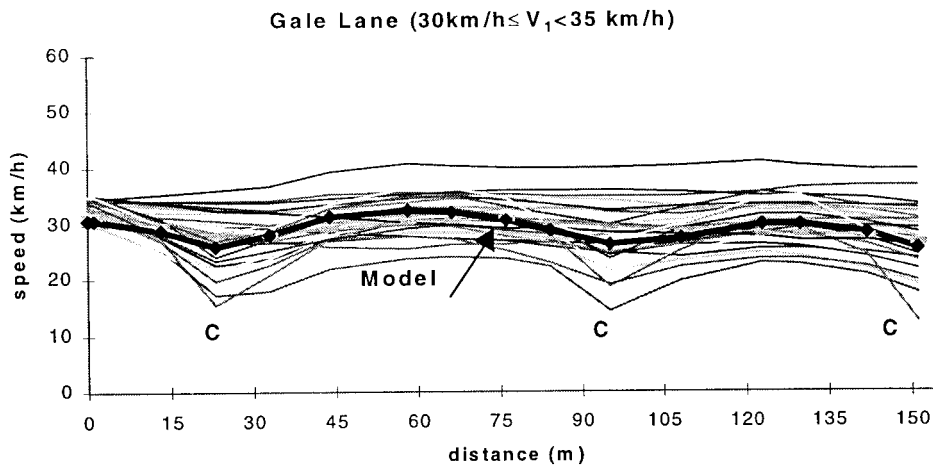


Fig. 14. Modelled speed profile ( $V_1 = 33\text{ km/h}$ ) vs observed speed profiles—Gale Lane.

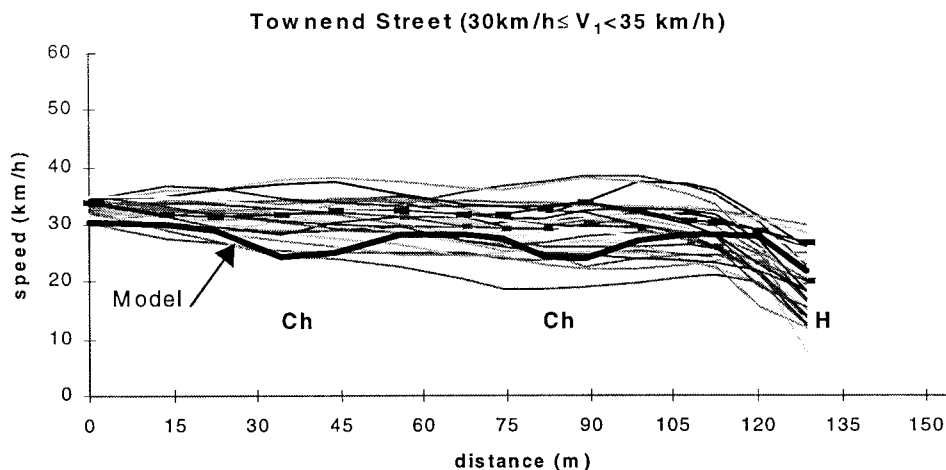


Fig. 15. Modelled speed profile ( $V_1 = 33\text{ km/h}$ ) vs observed speed profiles—Townend Street.

forces drivers to slow down, apparently to lower levels than the hump at the calibration site, Foxwood Lane West, and here the model is reasonably accurate.

Similarly to the analysis of calibration sites, speed profiles have also been plotted against modelled speed profiles for the average entry speed observed at each site as follows: Fourth Avenue, 22.39 km/h; Gale Lane, 29.89 km/h; and Townend Street, 30.25 km/h. Figs. 16–18 compare these predictions with the average of the observed data. In Fig. 16, the average and predicted profile curvatures for Fourth Avenue are similar, in contrast to the results for a 33 km/h input speed (Fig. 13). However, the model underpredicts values for much of the range, except at the cushion (the last data point). Gale Lane (Fig. 17) represents a good prediction. The model slightly underpredicts speeds by a constant value. The only exception appears at the acceleration length after the first cushion, before reaching maximum speed. This discrepancy seems to be responsible for the transference of the maximum speed data point to a previous point on the graph, also observed for Foxwood Lane East data. For Townend Street, the average profile shown in Fig. 18 reveals no impact on speeds from the first chicane and a slight impact from the second chicane.

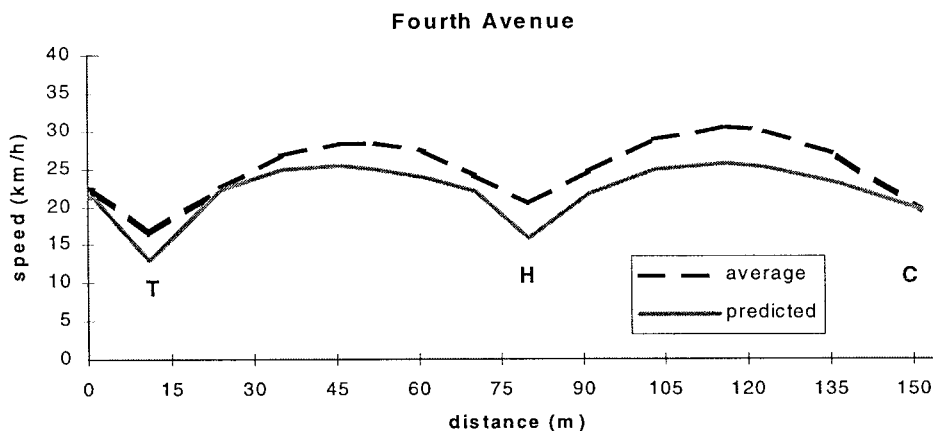


Fig. 16. Average vs modelled speed profile curve—Fourth Avenue.

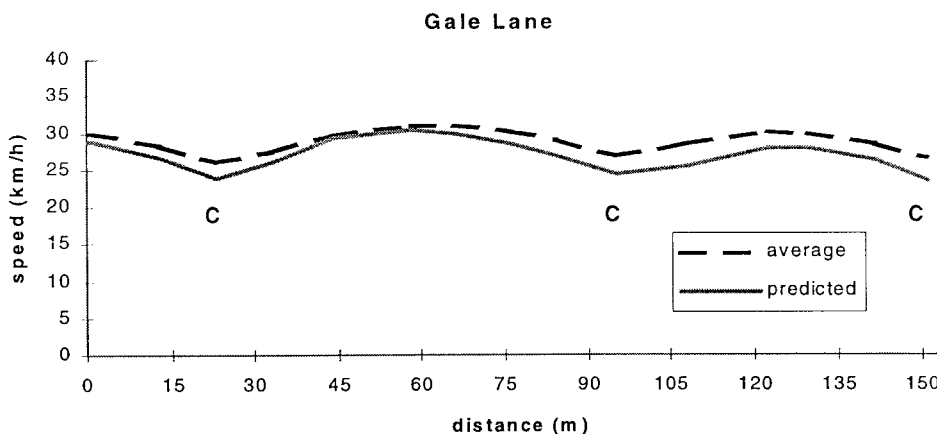


Fig. 17. Average vs modelled speed profile curve—Gale Lane.

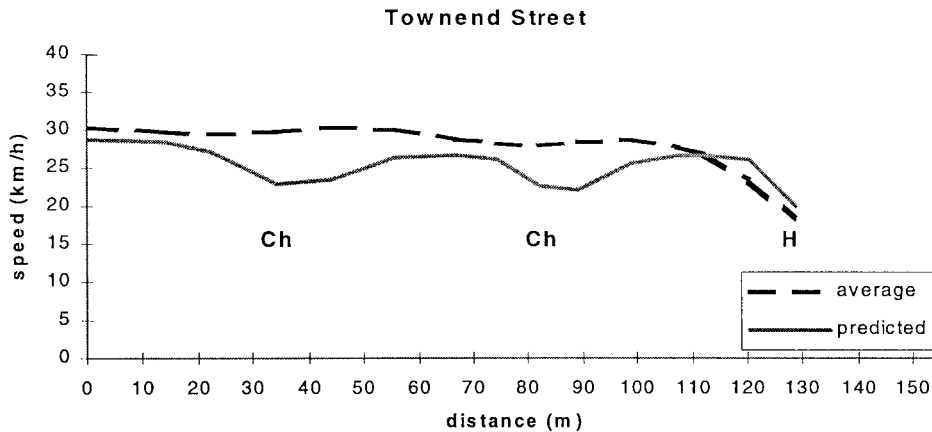


Fig. 18. Average vs modelled speed profile curve—Townend Street.

Not surprisingly the model fails to predict this lack of impact. Conversely, the hump induces a somewhat higher deceleration rate than that predicted by the model. This is probably because the model has assumed an impact of the chicanes which in practice does not occur.

In the calibration process, maximum speeds between measures have been predicted at a different data point from the average profile. Conversely, Fourth Avenue and Gale Lane presented a more realistic prediction for maximum speeds, except for the maximum speed between the first and second cushion in Gale Lane which appeared slightly earlier than the actual trends.

## 6. Conclusions

This examination of the effects of different types of traffic calming measures on speeds has shown differences by type of measure. The greatest impact on speeds was produced by tables, followed by humps, chicanes and cushions. Layfield (1994) has also shown the same order of impact for tables, humps and cushions. An indication of the relative impact of different measures is given through the coefficients for the dummy variables in the speed profile model.

The assessment of the speed distributions at the measures has indicated some differences between measures of the same type. The lower average speed at the cushion in Fourth Avenue may reflect its distinct dimensions in comparison to the remainder of the cushions. For both humps and tables average speeds are noticeably different when comparing calibration and validation sites (Figs. 5 and 19). As the speed differential between before and after speed measurements has been maintained, differences between measures of the same type may be mainly attributed to variations between sites in addition to the geometry and layout of the devices, for instance at Fourth Avenue and at Foxwood Lane. Fourth Avenue presents characteristics that differ from the remainder of the sites: low traffic volume and large trees all the way down the road which reduce the forward visibility and the perceived road width.

The model presented a reasonable predictive capability for the calibration sites as graphically attested. The superimposition of modelled curves on observed profiles revealed that the modelled curves showed a good correlation with the observed curves. Nevertheless, discrepancies in the

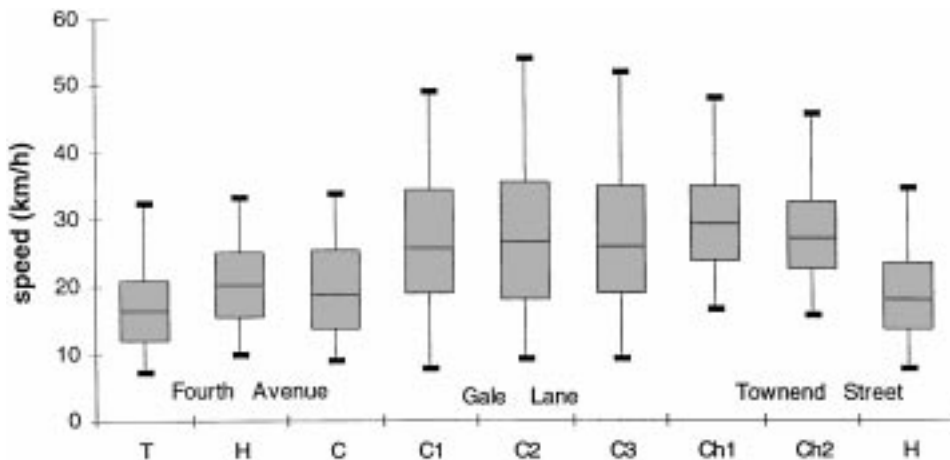


Fig. 19. Minimum, 15th percentile, median, 85th percentile and maximum speeds—validation sites.

entry speed prediction were observed, with only certain entry speed ranges proving satisfactory. Outside those ranges, results were overestimated for lower input speeds and underestimated for higher input speeds. Although all drivers did not attain maximum speed at the same data point, the predicted profile tended to underestimate distance to this point. These differences became even more noticeable in the acceleration profiles derived from the application of the speed profile model.

The model fits the validation data from Fourth Avenue and Gale Lane quite well, but it does not fit Townend Street largely because the chicanes there have no effect on speeds. The model response in terms of the entry speed range prediction presented discrepancies at all sites. As for the calibration sites, outside of certain speed ranges results were overestimated for lower input speeds and underestimated for higher input speeds. In general, the model has demonstrated its suitability for the validation sites except for Townend Street, hence indicating that chicanes cannot reliably be included in the model due to their wide design variation. The sample of chicanes used in this study demonstrated diverse speed reduction levels, therefore making the model prediction unrealistic for chicanes whose impact differs from those included in the calibration process.

Through the model validation and the sensitivity analysis, it appears that the speed profile model predicts more successfully when the input speed fed into the equation is close to the average value. Individual speed profiles suggested distinct behaviour according to the entry speed. In order to better reproduce that speed variability, it would be preferable to devise specific models pertaining to certain entry speed ranges so that there would be distinct models for lower, medium and high entry speeds.

Differences in average speed at the measures of the same type (e.g. hump and table) which were also observed before the implementation of the calming schemes, raise the issue of these differences being attributed to the site. The type of road may be of some importance to the model as it may control speeds within certain levels. The before speed at the site may be used to characterise the site and act as an explanatory variable for 'speed'. Alternatively the type of road could be described by dummy variables (e.g. arterial, secondary, residential).

Finally, in order to achieve more realistic predictions, it is desirable to restrict the type of measures to those considered in the development of the model. As previously mentioned, chicanes are a delicate issue and the model represents the impact of a chicane which has similar design to the first one in Livingstone Street. The difference in the impact of chicanes on speeds has raised the need for more in-depth analysis of their design. This study has produced evidence that some chicanes are totally ineffective in reducing speeds. This differs from the results of off-road trials conducted by Sayer and Parry (1994), but these should be analysed with care as they may not reproduce the on-road situation. Consequently, this issue still requires further study to represent the vehicle speeds on public roads. Further work is needed to improve the effectiveness of chicanes, and hence to predict their impacts more reliably.

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